PRIYANKA VERMA

Total no. of Pages: 03 *Roll no.* 2K20/ME/198

SIXTH SEMESTER

B.Tech

END TERM EXAMINATION **May-2023**

HU-302 Engineering Economics

Time: 03:00 Hours *Max. Marks: 50*

Note : All questions carry equal marks.
Answer any five questions
Assume suitable missing data, if any.

1.1 Discuss role of engineering and technology in achieving Sustainable Development Goal 2030. [5][CO#3]

1.2 a. The value of a machine depreciates from Rs. 16560 to Rs. 6245 in 6 years. What is the annual rate of depreciation? [2.5+2.5=5] [CO# 1]

b. A company hardware company purchases computer screen at a cost of Rs. 4500/screen. In case the company makes it, the fixed and variable cost would be Rs. 40,00,00/- and Rs. 2000/ screen respectively Should the manufacturer make or buy the screen, if the annual demand is of 1500 computers?

2.1 Write five salient features of the Indian economy. [5][CO# 1]

2.2 A person deposits Rs. 200 at the end of every month for 5 years in a saving bank account that pays interest @ 4% p.a. compounded monthly. Find the amount in his account at the end of 5 years. [5][CO#2]

3.1 What do you mean by Fiscal Policy? How companies can be encouraged to opt for green technology through using Fiscal Policy. Discuss. [2.5+2.5=5] [CO#1and3]

3.2 A company has to replace a machine in the production line after 11 years at the cost of Rs. 60,00,000/-. It plans to deposit an equal amount at the end of every year for next 11 years at an interest rate of 11 per cent which is compounded annually. Find the equivalent amount that must be deposited at the end of every year for the next 11 years. [5][CO#2]

4.1 What do you mean by Labour Intensive and Capital Intensive production process? Discuss with example. [5][CO#1]

4.2 A machine costs a company Rs. 10,000 and its expected life is 5 years. Alternatively, the machine can be obtained by leasing at the annual rent of Rs. 2500. If the rate of interest is 12% per annum, find which alternative is preferable to the company? [5][CO#2]

5.1 You are CEO of an Oligopolistic company. What are the factors you should consider while deciding price of your product. [5][CO#3]

5.2 Prepare a cost sheet showing the total cost and per tonne cost of paper manufactured by ABC paper mills Ltd. for the month of March 2023. There are 26 working days in the month. Also find the profit earned by the company. The details are as under: [5][CO# 2]

Direct Raw Material	Paper pulp	6000 tons @ Rs. 900/ton
Direct Labour Cost	Skilled workers	280@783.00/day
	Semi Skilled	300@ 712.00/ day
	Unskilled	470@646.00/ day
Direct Expenses	Equipment hire charges	Rs. 12000/ day
	Dyes	Rs. 650 per tonne of raw material input
Overheads	Variable	@ 50% of Direct Wages
	Fixed	Rs. 270000/month
Administrative overheads		12% of factory or work cost
Selling and Distribution overheads		20% of the prime cost
Profit		10% of the total cost

6.1 Discuss Business cycle. How engineers can help to bring the country back to the normal situation? Discuss [2.5+2.5=5] [CO#1and3]

6.2 Two years ago, a machine was purchased at a cost of Rs. 4,00,000 to be useful for ten years. Its salvage value at the end of its life is Rs. 60,000. The annual maintenance cost is Rs. 60,000. The market value of the present machine is Rs. 3,20,000. Now a new machine to cater to the need of the present machine is available at Rs. 3,80,000 to be useful for eight years. Its annual maintenance cost is Rs. [5] [CO#2]

45,000. The salvage value of the of the new machine is Rs. 40,000/- Using an interest rate of 11%, find whether it is worth replacing the present machine with new machine?

7.1 Discuss socio-economic factors which need to be considered while deciding foreign location of your company. [5][CO#3]

7.2 What amount of money saved today will yield Rs. 40,000 in the third year and Rs. 55,000/- after five years at the 12% rate of interest compounded annually. [5][CO#2]